

executable test suite has a test harness that links the test suite with the system under test (SUT) with the executable test suite enhancing proper communication between the test suite and the system under test (SUT).

What are the types of software testing?

Software testing can be broadly classified into two types:

- **Functional Testing:** Functional testing is done on a software application based on some predefined functional requirements. Each function of the system under test (SUT) is tested with suitable input, and the output acquired is validated against the predefined functional requirement. Functional testing can also be of many types:
 1. **Unit Testing:** In a complex and lengthy code, testing the entire code can give numerous errors that are difficult to debug. An easier way is to test every part or each unit of the code. Unit testing is testing the output of every unit or part of the code developed in a component. The entire code is well documented, having the errors of each part or unit of the code.
 2. **Component Testing:** Component testing is testing an entire component with suitable input to get an output which is then validated with the predefined and desired output. A component is any part that needs a suitable input to give us a suitable output to validate. It can be a sub-system of the entire system, a web page, or a code module.
 3. **Smoke Testing:** This is one of the essential types of functional testing that needs to be done at the very beginning. Smoke testing is testing the most important function or component of the system. Smoke testing does not test all the functions of a system exhaustively. The test cases chosen for smoke testing are the tests that verify the critical component or function of the system. This is done when a new design is sent for quality analysis to the quality analysis team. After successful completion of smoke testing, the other testing methods are subsequently completed. If the build does not pass the smoke testing successfully, then it is discarded and sent back to the design and development team. The problems or errors are removed, or a new build is created from scratch, which then again appears for the smoke testing. Once a build successfully qualifies the smoke testing, it is then put to other functional testing and regression testing for other functions of the system.
 4. **Integration Testing:** Integration testing is the most popular type of functionality testing. Integration testing is testing done on a group